Chapter 2: Comparing Two Population Means

We will compare **two population means** in this chapter. In particular, we will talk about:

The difference between **independent** and **dependent** samples,

Hypothesis tests and **confidence intervals** of means for two **independent samples** when $\sigma_1 = \sigma_2 = \sigma$ is **unknown**,

Hypothesis tests and **confidence intervals** of means for two **independent samples** when σ_1, σ_2 and are unknown and unequal.

Hypothesis tests and **confidence intervals** of means for **paired samples**.

How to calculate **necessary sample size** to reach target power.

Example 1: An experiment was performed on a manufacturing plant by making in sequence 10 batches of a chemical using a standard production method (A) followed by 10 batches using a modified method (B). The results are given in the following table:

A	B
89.7	84.7
81.4	86.1
84.5	83.3
84.8	91.9
87.3	86.3
79.7	79.3
85.1	82.6
81.7	89.1
83.7	83.7
84.5	88.5

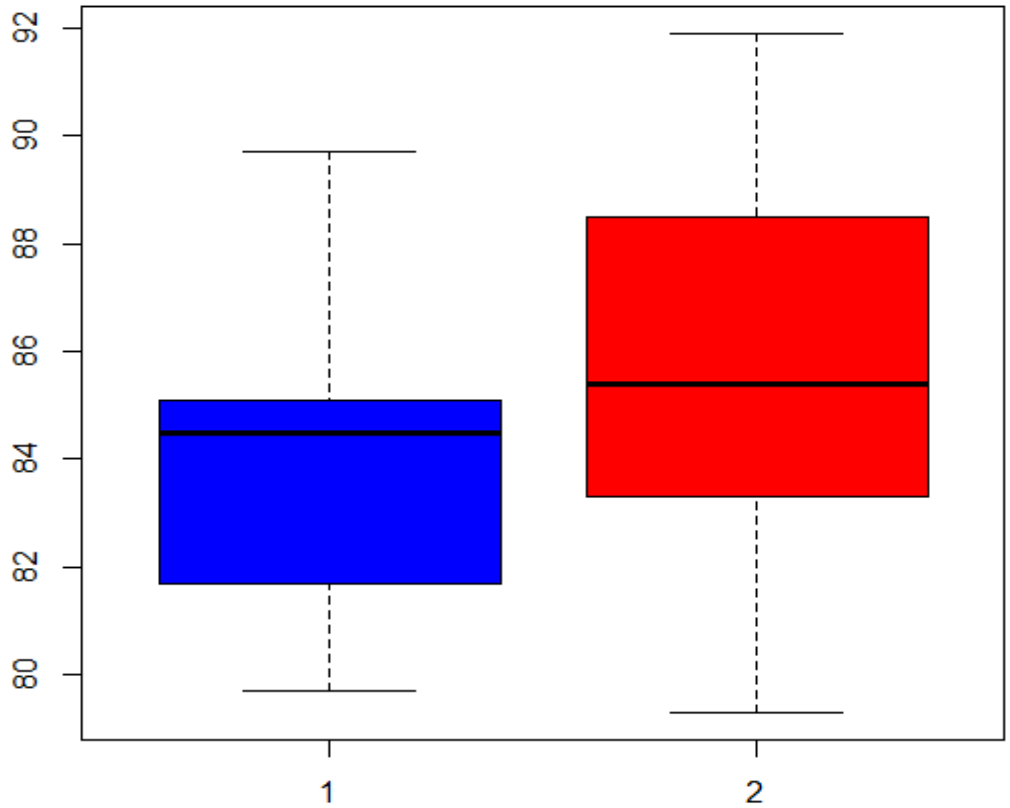
Do we have enough evidence to conclude that the method B gives **higher** yield than method A on **average**?

R Codes to Get Summary Statistics:

```
> A = c(89.7, 81.4, 84.5, 84.8, 87.3, 79.7, 85.1, 81.7, 83.7, 84.5)

> B = c(84.7, 86.1, 83.3, 91.9, 86.3, 79.3, 82.6, 89.1, 83.7, 88.5)

> summary(A)

   Min.   1st Qu.  Median    Mean  3rd Qu.    Max.
  79.70   82.20    84.50    84.24   85.02    89.70

> summary(B)

   Min.   1st Qu.  Median    Mean   3rd Qu.   Max.
   79.30   83.40   85.40    85.55   87.95    91.90
```

R Code to Get Boxplot:

```
> boxplot(A,B, col=c("blue","red"))
```



Example 1 (Cont.): Do we have enough evidence to conclude that the method B gives **higher** yield than method A on **average**?

$$\mu_B > \mu_A$$

$$\mu_A - \mu_B < 0$$

OR

$$\mu_B - \mu_A > 0$$

We are going to test

$$H_0 : \mu_A - \mu_B = 0 \text{ vs. } H_a : \mu_A - \mu_B < 0$$

The value of test statistic is:

$$TS = \frac{\text{Estimate} - \text{Null Hypothesis}}{\text{S.E.(Estimate)}}$$

$$= \frac{\bar{y}_A - \bar{y}_B - 0}{\text{S.E.}(\bar{y}_A - \bar{y}_B)} \sim t - \text{distribution}$$

Recall that, for two **independent** samples, We have

$$SD(\bar{y}_1 - \bar{y}_2) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

If we assume $\boldsymbol{\sigma_1 = \sigma_2 = \sigma}$, then the above standard deviation can be re-written as:

$$SD(\bar{y}_1 - \bar{y}_2) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} =$$

$$= \sqrt{\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}} = \sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

We still don't know the true **common standard deviations** of the two groups, so we need to estimate and use the standard error

$$S.E.(\bar{y}_A - \bar{y}_B) = s_{\text{Pooled}}\sqrt{\frac{1}{n_A} + \frac{1}{n_B}}$$

Question: what is s_{Pooled}?

$$S.E.(\overline{y}_A - \overline{y}_B) = s_{\text{Pooled}}\sqrt{\frac{1}{n_A} + \frac{1}{n_B}}$$

Question: Should we use s_1 or s_2 to estimate σ?

$$n_A = 10, \overline{y}_A = 84.24, s_A = 2.9018$$

and

$$n_B = 10, \overline{y}_B = 85.55, s_B = 3.6433$$

$$s_p = \sqrt{\frac{(n_A - 1)s_1^2 + (n_B - 1)s_2^2}{n_A + n_B - 2}}$$

$$= \sqrt{\frac{(10-1)1(2.9018)^2 + (10-1)(3.6433)^2}{10 + 10 - 2}} = 3.2935$$

It can be shown that $s_p = \sqrt{\frac{s_1^2 + s_2^2}{2}}$ if $n_1 = n_2$

Example 1 (Cont.):

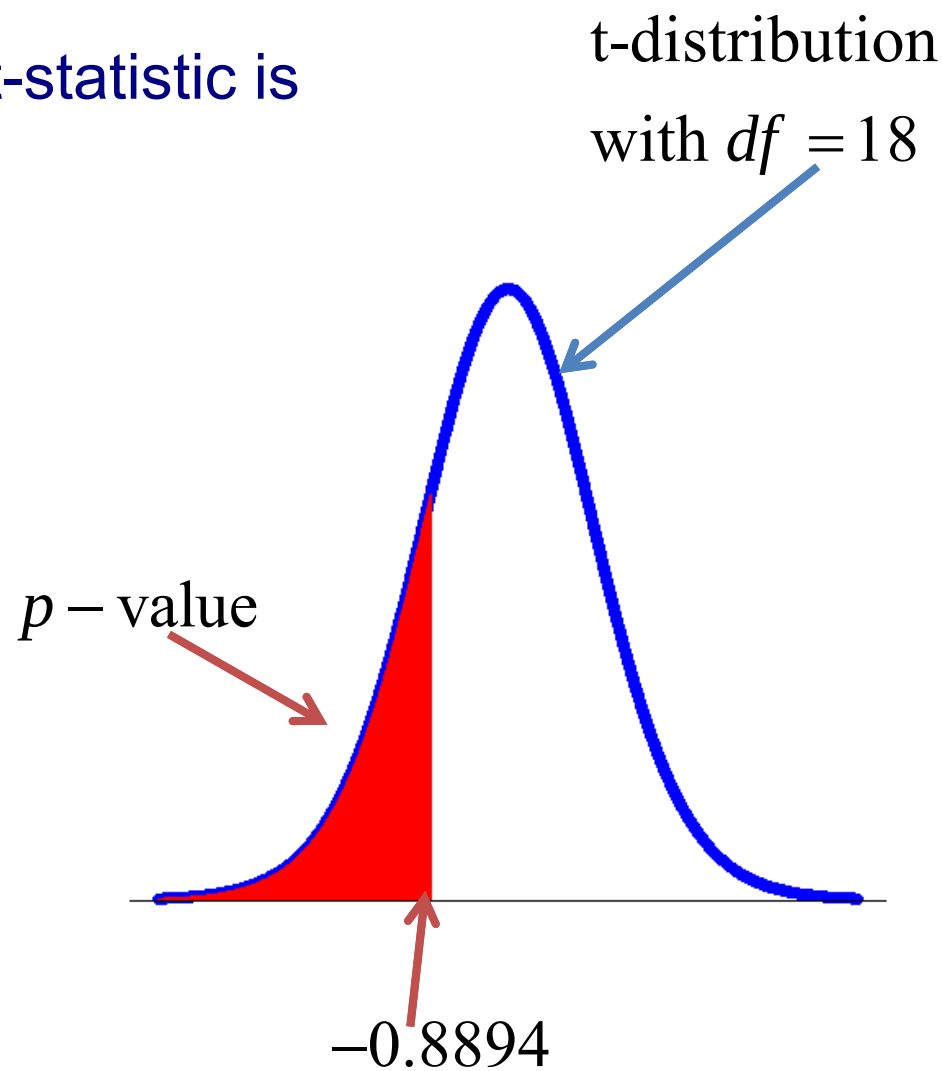
Goal: Testing $H_0 : \mu_A - \mu_B = 0$ vs. $H_a : \mu_A - \mu_B < 0$

Test-statistic: The value of test-statistic is

$$TS = t - \text{test} = \frac{\bar{y}_1 - \bar{y}_2 - 0}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$= \frac{84.24 - 85.55 - 0}{3.2935\sqrt{\frac{1}{10} + \frac{1}{10}}} = -0.8894$$

P-value: P(t_{18}<−0.8894)



Cutoff probabilities

degrees of freedom	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
1	1.000	1.376	1.963	3.078	6.314	12.71	15.89	31.82	63.66	127.3	318.3	636.6
2	0.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.09	22.33	31.60
3	0.765	.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.21	12.92
4	0.741	.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	0.727	.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	0.718	.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	0.711	.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	0.706	.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	0.703	.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	0.700	.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	0.697	.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	0.695	.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	0.694	.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	0.692	.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	0.691	.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073
16	0.690	.865	1.071	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015
17	0.689	.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	0.688	.862	1.067	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.611	3.922
19	0.688	.861	1.066	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	0.687	.860	1.064	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850
21	0.663	.858	1.061	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819
22	0.686	.858	1.061	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792

Example 1 (Cont.): Find a 95% confidence interval for the **difference** of two **population means**.

A 95% confidence interval for $\mu_A - \mu_B$ is:

$$\text{Estimate} \pm \text{Critical Value} \times \text{S.E.(Estimate)}$$

$$\bar{y}_A - \bar{y}_B \pm t^* s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} =$$

$$84.24 - 85.55 \pm (2.101)\left(3.2935\sqrt{\frac{1}{10} + \frac{1}{10}}\right)$$

$$-1.3 \pm 3.095 \Rightarrow (-4.395, 1.795)$$

R Codes to Perform Two Sample T-test (With Equal Variance Assumption – One-sided Alternative):

```
> t.test(A, B, alternative =  "less", var.equal =T)


        Two Sample t-test

data:  A and B
t = -0.8894, df = 18, p-value = 0.1928
alternative hypothesis: true difference in means is less than 0
95 percent confidence interval:
     -Inf 1.244102
sample estimates:
mean of x mean of y
   84.24     85.55
```

R Codes to Perform Two Sample T-test (With Equal Variance Assumption – Two-sided Alternative):

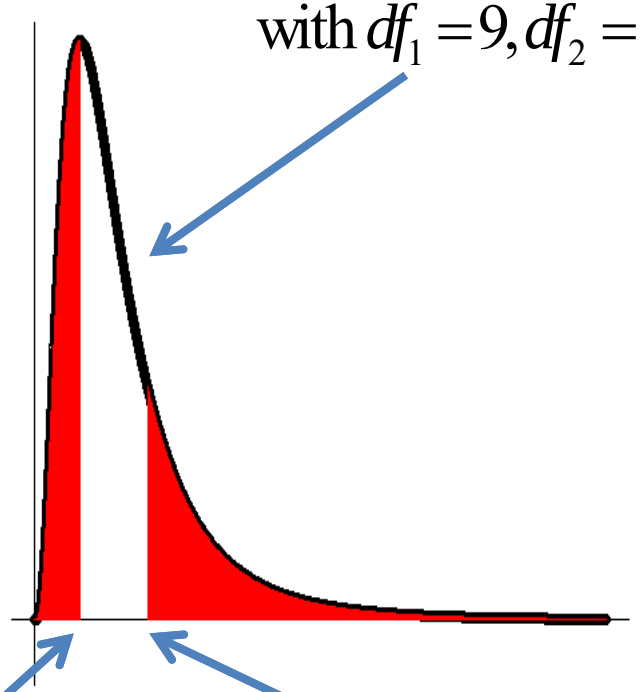
```
> t.test(A, B, var.equal =T)

 Two Sample t-test

data:  A and B
t = -0.8894, df = 18, p-value = 0.3855
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -4.404447  1.784447
sample estimates:
mean of x mean of y
   84.24     85.55
```

Check Equal Variance Assumption:

The value test statistic for testing H_0: $\sigma_1 = \sigma_2 \rightarrow \frac{\sigma_1^2}{\sigma_2^2} = 1$ is:

$$TS = F - \text{test} = \frac{s_1^2}{s_2^2} = \frac{2.9018^2}{3.6433^2} = 0.6344$$

The p-value is 2P($F_9^9 < 0.6344$)



$0.6344 \quad 1/0.6344 = 1.5763$

		1	2	3	4	5	6	7	8	9	10
8	0.25	1.54	1.66	1.67	1.66	1.66	1.65	1.64	1.64	1.63	1.63
8	0.1	3.46	3.11	2.92	2.81	2.73	2.67	2.62	2.59	2.56	2.54
8	0.05	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35
8	0.025	7.57	6.06	5.42	5.05	4.82	4.65	4.53	4.43	4.36	4.30
8	0.01	11.26	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91	5.81
8	0.005	14.69	11.04	9.60	8.81	8.30	7.95	7.69	7.50	7.34	7.21
8	0.001	25.41	18.49	15.83	14.39	13.48	12.86	12.40	12.05	11.77	11.54
		1	2	3	4	5	6	7	8	9	10
9	0.25	1.51	1.62	1.63	1.63	1.62	1.61	1.60	1.60	1.59	1.59
9	0.1	3.36	3.01	2.81	2.69	2.61	2.55	2.51	2.47	2.44	2.42
9	0.05	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14
9	0.025	7.21	5.71	5.08	4.72	4.48	4.32	4.20	4.10	4.03	3.96
9	0.01	10.56	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35	5.26
9	0.005	13.61	10.11	8.72	7.96	7.47	7.13	6.88	6.69	6.54	6.42
9	0.001	22.86	16.39	13.90	12.56	11.71	11.13	10.70	10.37	10.11	9.89

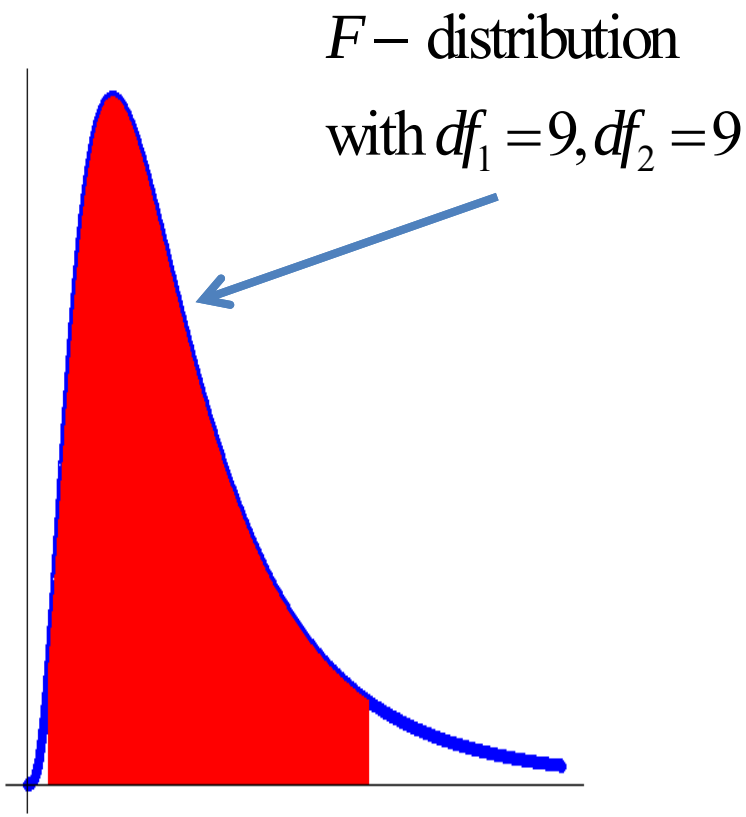
Check Equal Variance Assumption:

A 95% confidence interval for $\frac{\sigma_1^2}{\sigma_2^2}$ is:

$$\frac{1}{F_{\alpha/2}(n_1-1,n_2-1)}\frac{s_1^2}{s_2^2} < \frac{\sigma_1^2}{\sigma_2^2} < F_{\alpha/2}(n_2-1,n_1-1)\frac{s_1^2}{s_2^2}$$

$$\Rightarrow \frac{1}{4.03}\times\frac{2.9018^2}{3.6433^2} < \frac{\sigma_1^2}{\sigma_2^2} < 4.03\times\frac{2.9018^2}{3.6433^2}$$

$$\Rightarrow 0.157 < \frac{\sigma_1^2}{\sigma_2^2} < 2.55$$

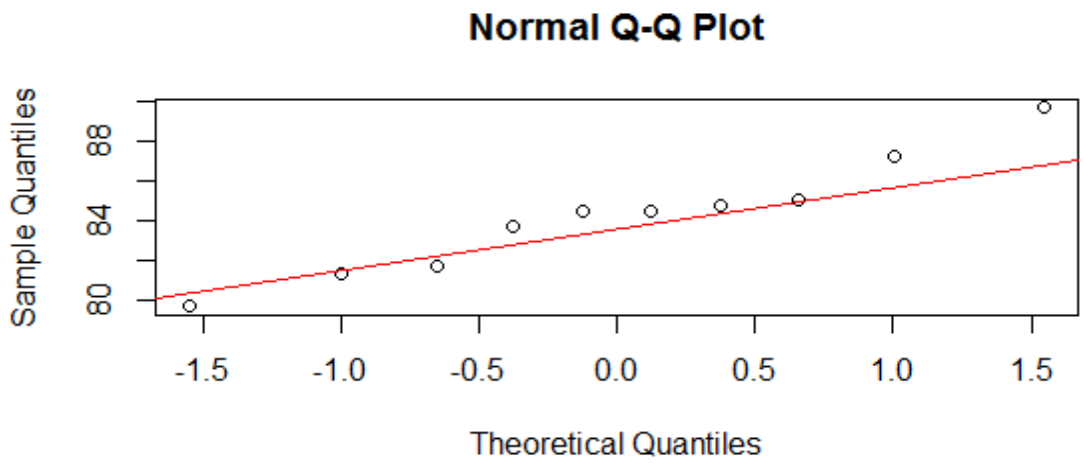


R Code to Check Equal Variance Assumption:

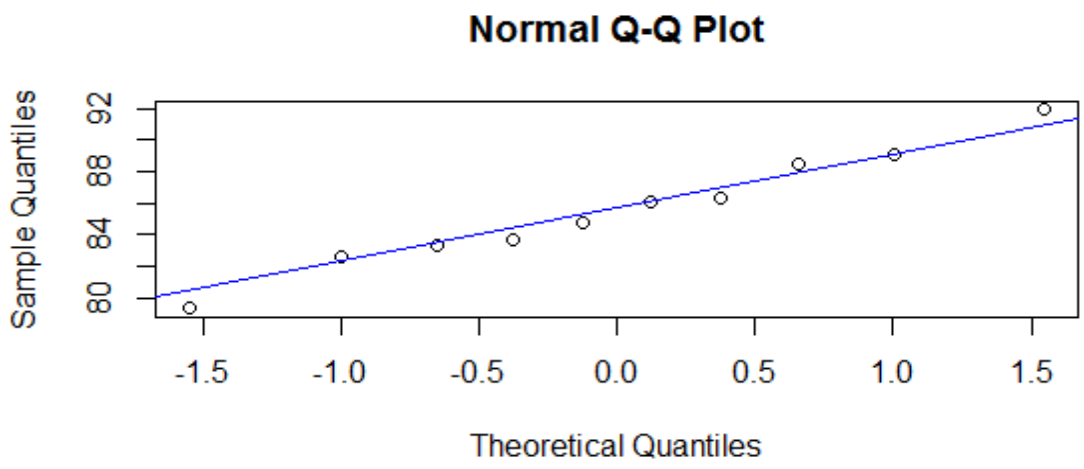
```
> var.test(A,B)

 F test to compare two variances

data:  A and B
F = 0.6344, num df = 9, denom df = 9, p-value = 0.5084
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.1575664     2.5539358
sample estimates:
ratio of variances
        0.6343615
```

R Codes to Check Normality Assumption:

```
>par(mfrow =c(2,1))

> qqnorm(A)
> qqline(A, col=2)
```




```
> qqnorm(B)
> qqline(b, col=4)
```



Two Sample t-test (with unequal variance assumption)

The test statistic for testing H_0: $\mu_1 - \mu_2 = 0$ is

$$TS = t - \text{test} = \frac{\bar{y}_1 - \bar{y}_2 - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

The null distribution or the distribution of test-statistic under null hypothesis when the null hypothesis is true can be **approximated** by a t-distribution with

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(s_1^2 / n_1\right)^2}{n_1 - 1} + \frac{\left(s_2^2 / n_2\right)^2}{n_2 - 1}}$$

Example 1 (Cont.):

Goal: Testing $H_0 : \mu_A - \mu_B = 0$ vs. $H_a : \mu_A - \mu_B < 0$

Test-statistic: The value of test-statistic is

$$TS = t - \text{test} = \frac{\bar{y}_1 - \bar{y}_2 - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$= \frac{84.24 - 85.55}{\sqrt{\frac{2.9018^2}{9} + \frac{3.6433^2}{9}}}$$

$$= -0.8894$$

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(s_1^2 / n_1\right)^2}{n_1 - 1} + \frac{\left(s_2^2 / n_2\right)^2}{n_2 - 1}}$$

$$= \frac{\left(\frac{2.9018^2}{10} + \frac{3.6433^2}{10}\right)^2}{\frac{(2.9018^2 / 10)^2}{9} + \frac{(3.6433^2 / 10)^2}{9}}$$

$$= 17.142$$

P-value: P($t_{17.142}$<−0.8894)

R Code to Perform Two Sample t-test (with unequal variance assumption - Welsh Approximation):

```
> t.test(A, B, alternative =  "less")

        Welch Two Sample t-test

data:  A and B
t = -0.8894, df = 17.142, p-value = 0.193
alternative hypothesis: true difference in means is less than 0
95 percent confidence interval:
     -Inf 1.251045
sample estimates:
mean of x mean of y
   84.24     85.55
```

Example 2: The blood pressure of five patients were recorded before and 30 minutes after receiving a drug. The results are given in the following table.

Patients	1	2	3	4	5	Mean	S.D.
Before	146	134	110	126	157	134.6	18.105
After	140	138	110	116	144	129.6	13.822

Is the drug useful in **reducing** the blood pressure?

Note:

1) It is about comparing **two population means**.

2) Two samples are **NOT independent**.

3) This is paired samples.

Example 1 (Cont.): Is the drug useful in **reducing** the blood pressure?

$$\mu_{\text{Before}} > \mu_{\text{After}}$$

$$\mu_{\text{Before}} - \mu_{\text{After}} > 0 \qquad \text{OR} \qquad \mu_{\text{After}} - \mu_{\text{Before}} < 0$$

Goal: $H_0 : \mu_{\text{Before}} - \mu_{\text{After}} = 0$ vs. $H_a : \mu_{\text{Before}} - \mu_{\text{After}} > 0$

Interested parameter: $\mu_{\text{Before}} - \mu_{\text{After}}$

Estimate: $\bar{y}_{\text{Before}} - \bar{y}_{\text{After}} = 134.6 - 129.6 = 5$

Note that $SD(\bar{y}_{Before} - \bar{y}_{After})$ is **NOT** equal to $\sqrt{\frac{\sigma^2_{Before}}{n_{Before}} + \frac{\sigma^2_{After}}{n_{After}}}$,

since data before and after are not independent.

Patients	1	2	3	4	5	Mean	S.D.
Before	146	134	110	126	157	134.6	18.105
After	140	138	110	116	144	129.6	13.822
Difference	6	-4	0	10	13	5.0	7.000

Define $d = Before - After$, then $\bar{d} = 5$, but

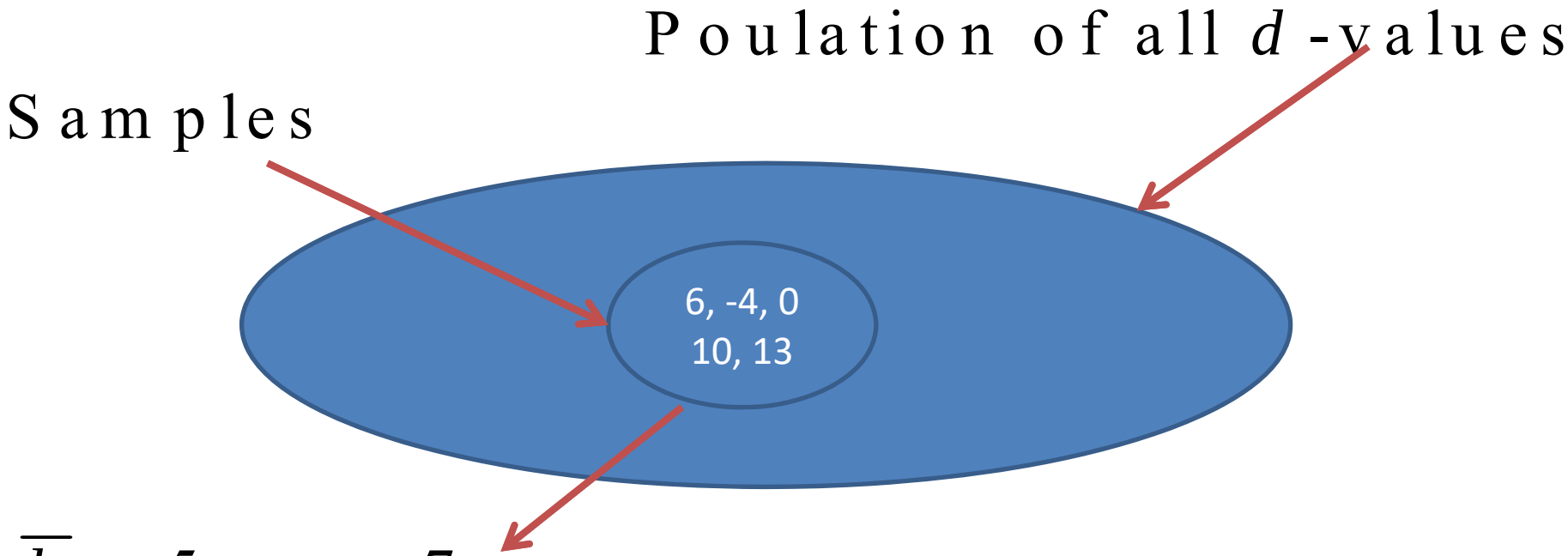
$$\bar{y}_{\text{Before}} - \bar{y}_{\text{After}} = 134.6 - 129.6 = 5$$

$$s_d = \sqrt{\frac{\sum (d - \bar{d})^2}{n-1}} = 7$$

Standard deviation of d-values

Therefore, $\bar{y}_{\text{Before}} - \bar{y}_{\text{After}} = \bar{d}$

$$\text{S.E.}(\bar{y}_{\text{Before}} - \bar{y}_{\text{After}}) = S.E.(\bar{d}) = \frac{s_d}{\sqrt{n}} = \frac{7}{\sqrt{5}} = 3.13$$



$\bar{d} = 5, s_d = 7$

$$H_0 : \mu_d = \mu_{Before} - \mu_{After} = 0$$

Necessary assumptions: d-values are independent random samples from a normal population or sample size is greater than 30.

$$TS = \frac{\bar{d} - 0}{s_d / \sqrt{n}} = \frac{5 - 0}{7 / \sqrt{5}} = 1.60$$

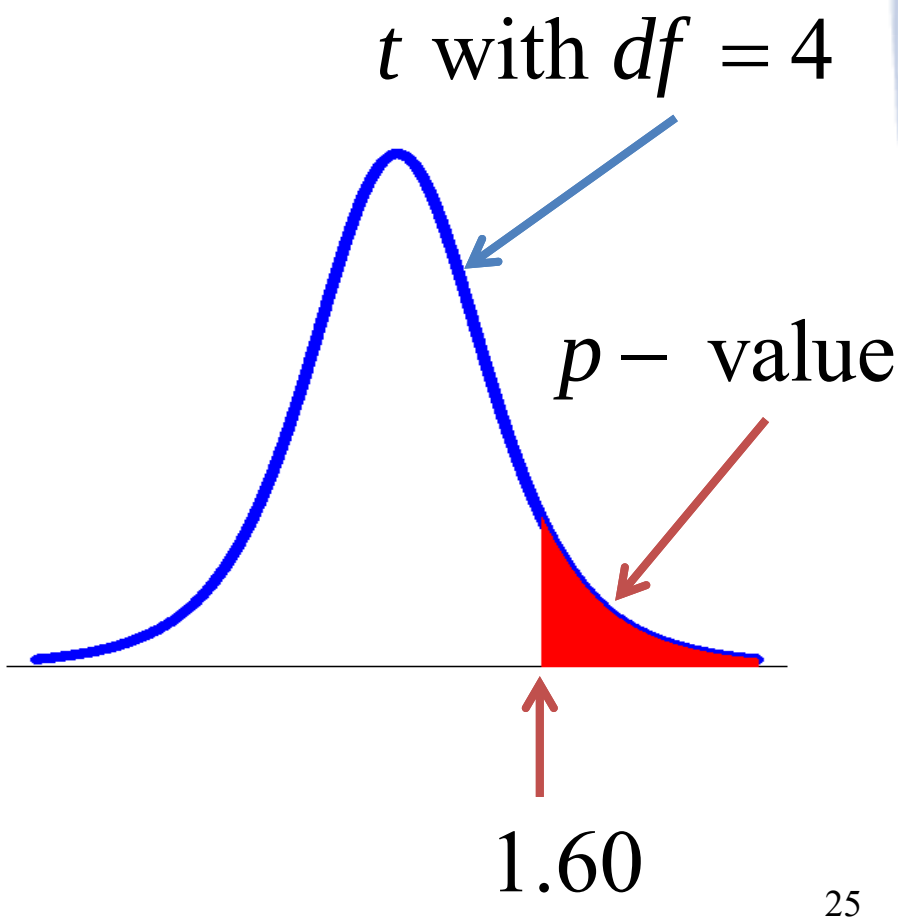
Example 1 (Cont.): Is the drug useful in **reducing** the blood pressure?

Goal: $H_0 : \mu_{\text{Before}} - \mu_{\text{After}} = 0$ vs. $H_a : \mu_{\text{Before}} - \mu_{\text{After}} > 0$

Test statistic:

$$TS = \frac{\bar{d} - 0}{s_d / \sqrt{n}} = \frac{5 - 0}{7 / \sqrt{5}} = 1.60$$

P-value: $P(TS > 1.60)$



Cutoff probabilities

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1	1.000	1.376	1.963	3.078	6.314	12.71	15.89	31.82	63.66	127.3	318.3	636.6
2	0.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.09	22.33	31.60
3	0.765	.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.21	12.92
4	0.741	.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	0.727	.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	0.718	.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	0.711	.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	0.706	.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	0.703	.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	0.700	.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	0.697	.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	0.695	.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	0.694	.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	0.692	.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	0.691	.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073
16	0.690	.865	1.071	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015
17	0.689	.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	0.688	.862	1.067	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.611	3.922
19	0.688	.861	1.066	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	0.687	.860	1.064	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850
21	0.663	.858	1.061	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819
22	0.686	.858	1.061	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792

Example 1 (Cont.): Find a 95% confidence interval for the **difference** of two **population means**.

A 95% confidence interval for $\mu_{Before} - \mu_{After}$ is:

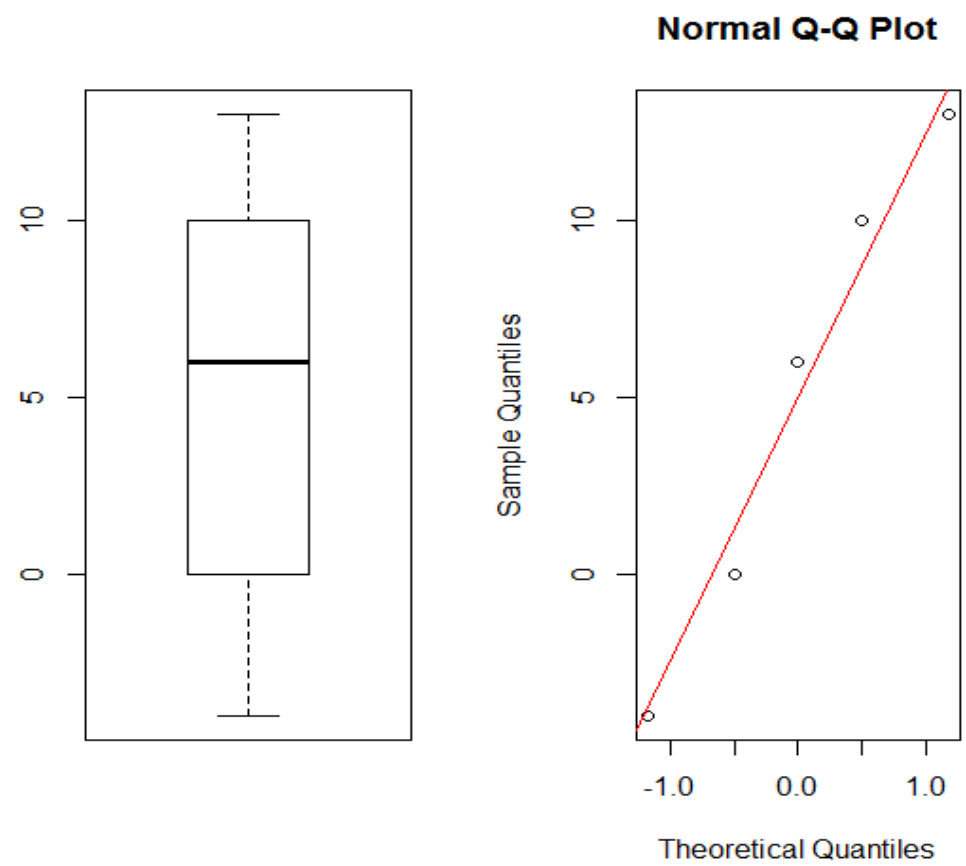
$$\text{Estimate} \pm \text{Critical Value} \times \text{S.E.(Estimate)}$$

$$\bar{d} \pm t^* \frac{s_d}{\sqrt{n}}$$

$$5 \pm (2.776)\frac{7}{\sqrt{5}} = 5 \pm 8.7 \Rightarrow (-3.7, 13.7)$$

R Codes to Get Boxplot and to Check Normality Assumptions:

```
> B = c(146, 134, 110, 126,157)

> A =c(140, 138, 110, 116, 144)

>d=B-A

>par(mfrow =c(1,2))

>boxplot(d)

> qqnorm(d)

> qqline(d, col=2)
```



R Codes to Perform T-test for paired Samples:

```
> t.test(B, A, alternative =  "greater", paired =T)

 Paired t-test

data:  B and A
t = 1.5972, df = 4, p-value = 0.09273
alternative hypothesis: true difference in means is greater than 0
95 percent confidence interval:
 -1.673736       Inf
sample estimates:
mean of the differences
                      5
```

Samples Size Calculations: The power of a test is the probability of rejecting the null hypothesis when it is false. Suppose:

1) we would like a power of at least .90 when the mean difference is $\delta = \mu_1 - \mu_2$ is 2.

2) $\alpha = 0.05,$ that is we reject H_0 if p-value is less than 0.05

3) Estimate σ. Since $s_p = 3.2935$, let use $\hat{\sigma} = 4$.

You can use the following command in R to calculate necessary sample size to get target power.

```
> power.t.test(delta = 2, sd = 4,sig.level = .05,power = .90,type =
"two.sample", alternative = "two.sided")
```

Samples Size Calculations:

```
> power.t.test(delta = 2, sd = 4,sig.level = .05,power = .90,type =
"two.sample", alternative = "two.sided")
```

Two-sample t test power calculation

```
          n = 85.03129
      delta = 2
         sd = 4
  sig.level = 0.05
      power = 0.9
alternative = two.sided
```

NOTE: n is number in *each* group

What Have We Learned?

Two-sample **t-methods** allow us to draw conclusion about the **difference** between the **means** of **two quantitative populations**, based on **independent samples**.

Parameter: $\mu_1 - \mu_2$

Estimate: $\bar{y}_1 - \bar{y}_2$

There are two t-methods: **pooled t-methods** and **t-methods without pooling**.

If we assume $\sigma_1 = \sigma_2$, then we use **pooled t-methods** .Otherwise, we use **t-methods without pooling**.

What Have We Learned (Cont.)?

t-methods without pooling:

Null Hypothesis: $H_0: \mu_1 - \mu_2 = 0$

Test Statistic: $\mathrm{TS} = t - test = t_o = \dfrac{\bar{y}_1 - \bar{y}_2 - 0}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}}$

Confidence Interval

$$\bar{y}_1 - \bar{y}_2 \pm t^* \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

$$df = \frac{\left(\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}\right)^2}{\dfrac{\left(s_1^2 / n_1\right)^2}{n_1 - 1} + \dfrac{\left(s_2^2 / n_2\right)^2}{n_2 - 1}}$$

What Have We Learned (Cont.)?

Pooled t-methods:

Null Hypothesis: $H_0: \mu_1 - \mu_2 = 0$

Test Statistic: $\mathrm{TS} = t - test = t_o = \dfrac{\bar{y}_1 - \bar{y}_2 - 0}{s_p\sqrt{\dfrac{1}{n_1} + \dfrac{1}{n_2}}}$ $\quad df = n_1 + n_2 - 2$

Confidence Interval

$$\bar{y}_1 - \bar{y}_2 \pm t^* \, s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$\text{where } s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

What Have We Learned (Cont.)?

Data are **paired** when the observations are collected in pairs or the observations in one group are related to observations in other. The simplest form of pairing is when we measure each subject twice – often before and after a treatment is applied.

Parameter: $\mu_{d=}\mu_1 - \mu_2$

Estimate: $\bar{d} = \bar{y}_1 - \bar{y}_2$

Test statistic: $\dfrac{\bar{d} - 0}{s_d / \sqrt{n}}$

Confidence Interval:

$$\bar{d} \pm t^* \frac{s_d}{\sqrt{n}}, \quad \text{where } df = n - 1$$

What Have We Learned (Cont.)?

We can use **power.t.test** in R to calculate necessary sample size to get target power. Use **help(power.t.test)** in R to get more details on this function!!!.